Task 1 Neural Network

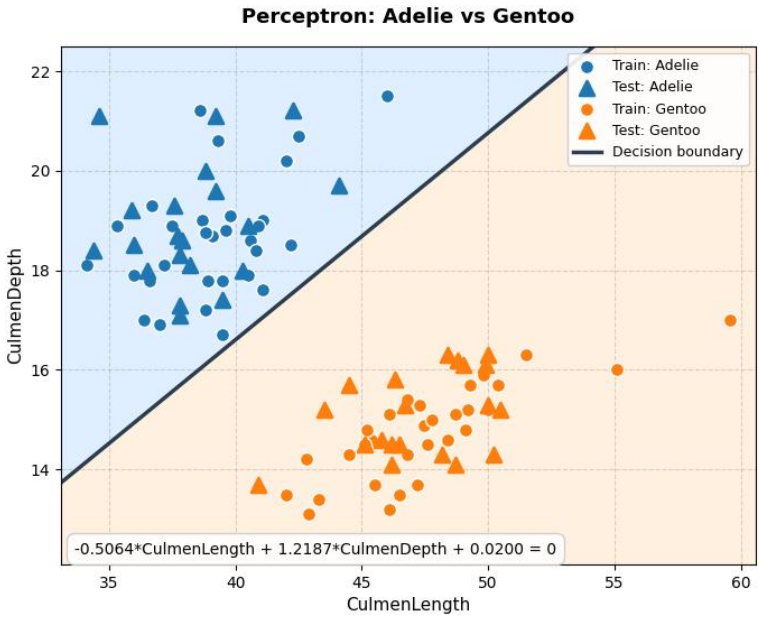
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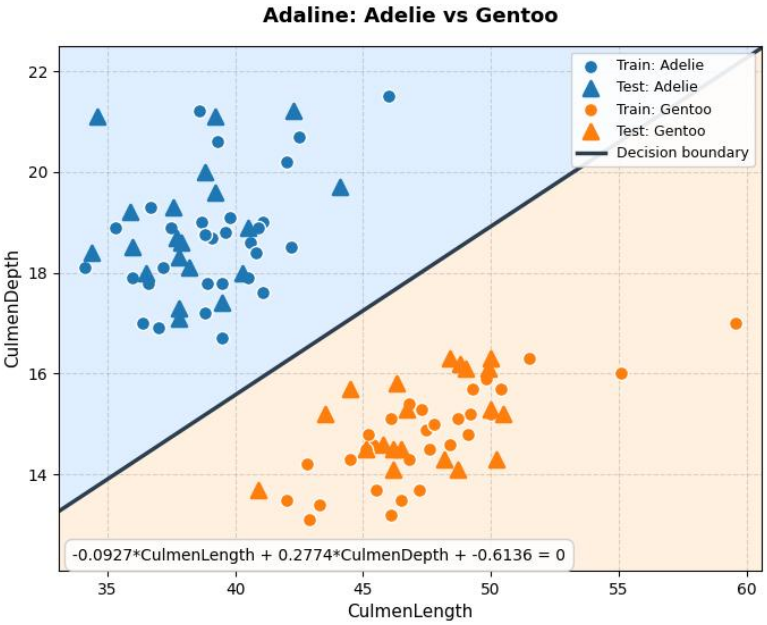
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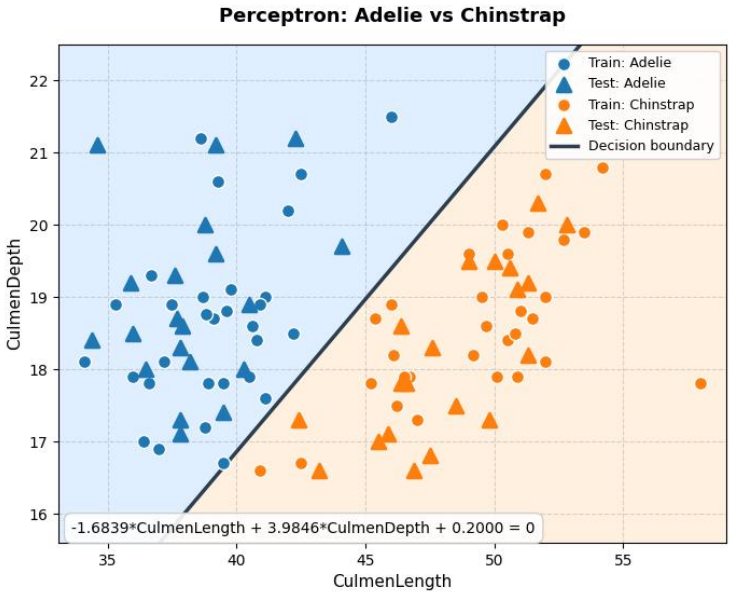
Introduction

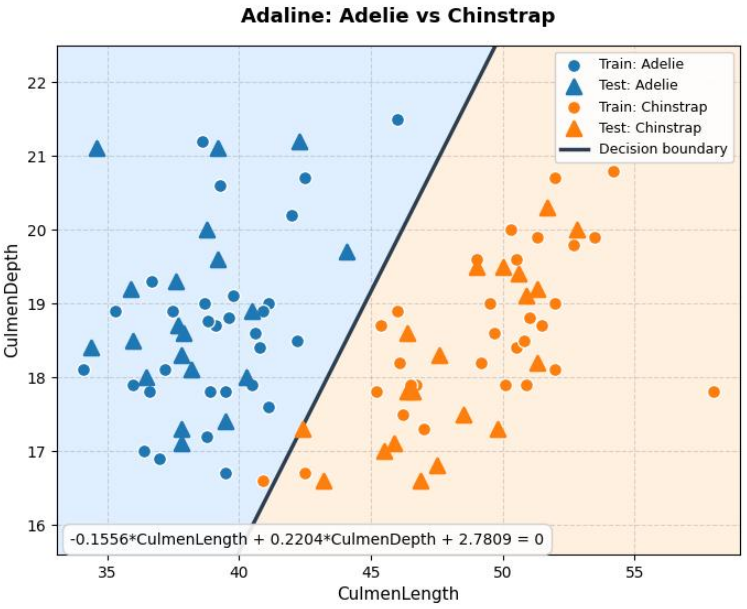
We are comparing Perceptron and Adaline using the penguins dataset. We picked 5 combinations of features to show what happens when data is easy (separable) and hard (overlapping)

Good Separable Combinations





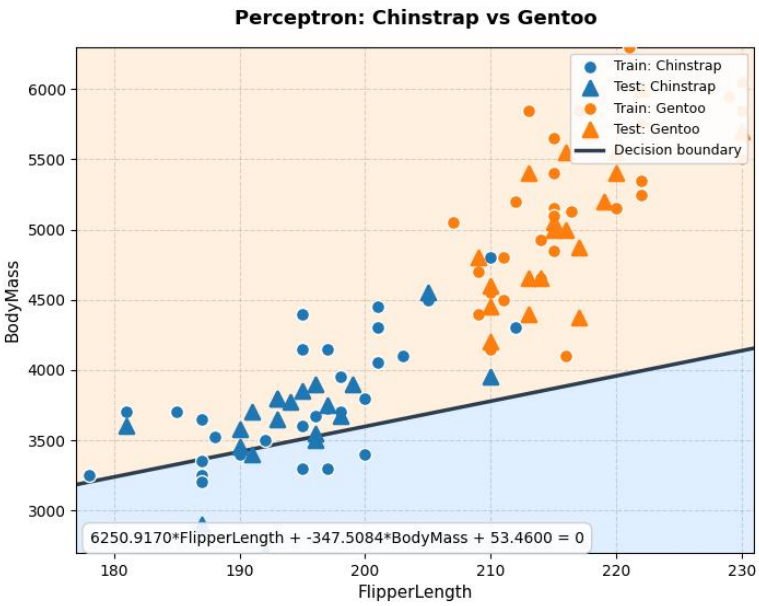


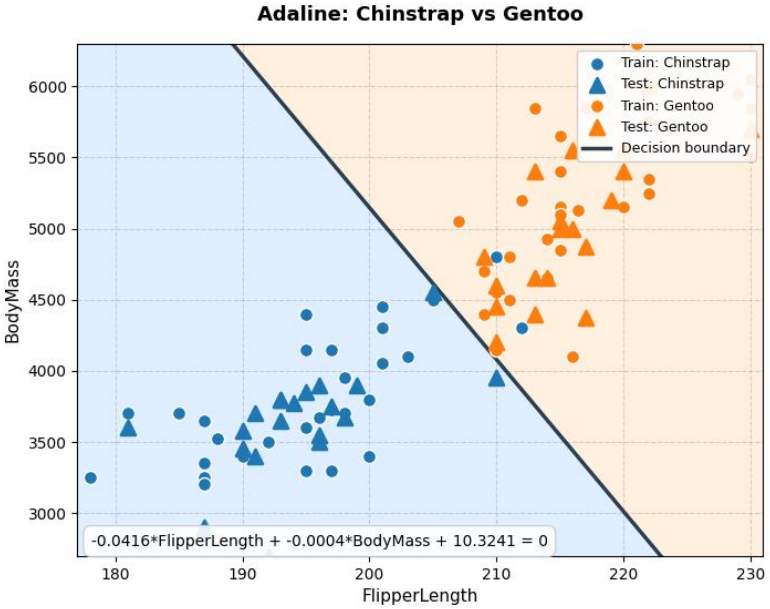


Conclusion:

When the features give a clear gap between the penguins, both Perceptron and Adaline get 100% accuracy. They just draw a line in the empty space. This proves both algorithms work perfectly on linearly separable data

Weird Case

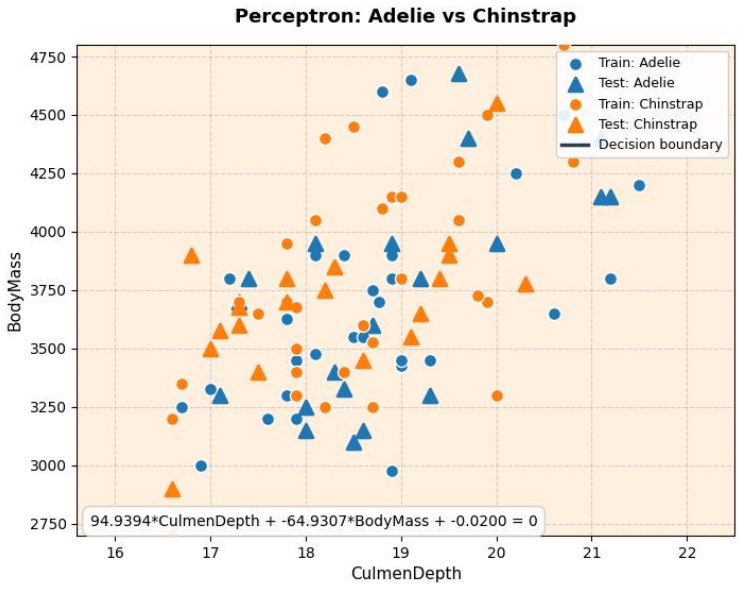


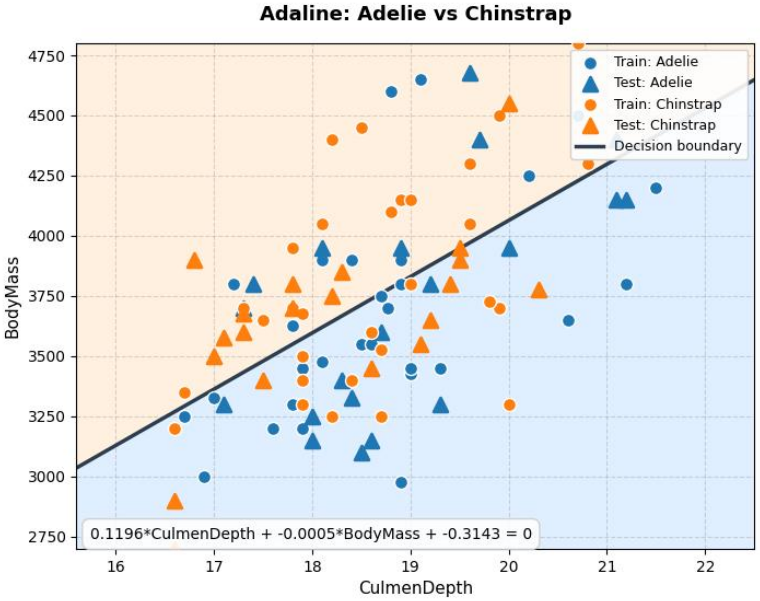


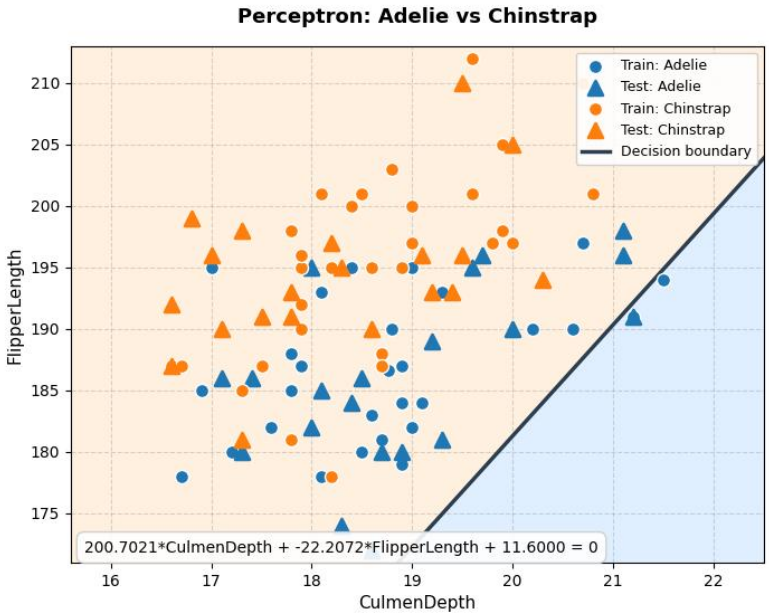
Conclusion:

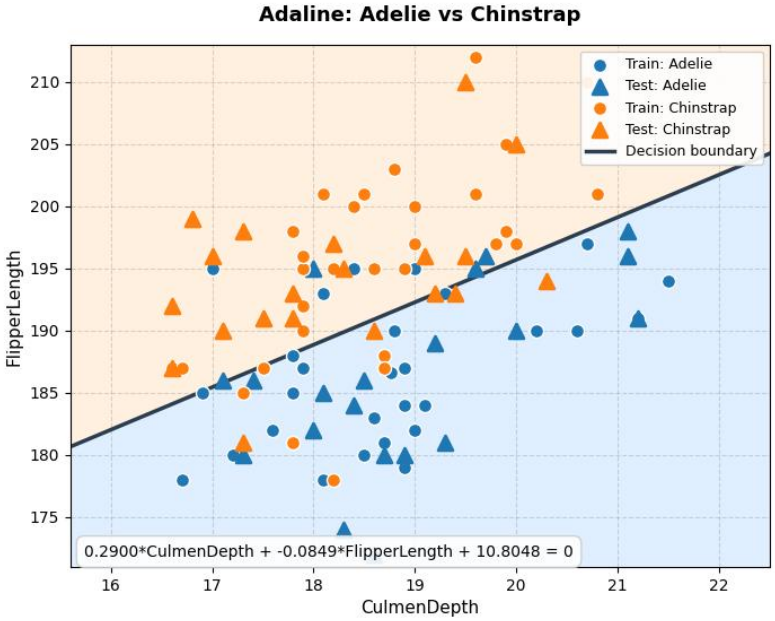
Even when data is separate, Perceptron sometimes gets stuck or draws a bad line if the learning rate or scales are off (it only got 60%). Adaline uses mean squared error to keep adjusting, so it found the perfect line (100%). This shows Adaline is more reliable

Bad and Mediocre Separable Combinations









Conclusion:

When the penguin features overlap, a straight line can't separate them. Perceptron totally freaks out because it never stops trying to find a perfect line that doesn't exist. Adaline is smarter because it just tries to find the "best possible" line with the lowest error, even if it's not perfect

Table Results:

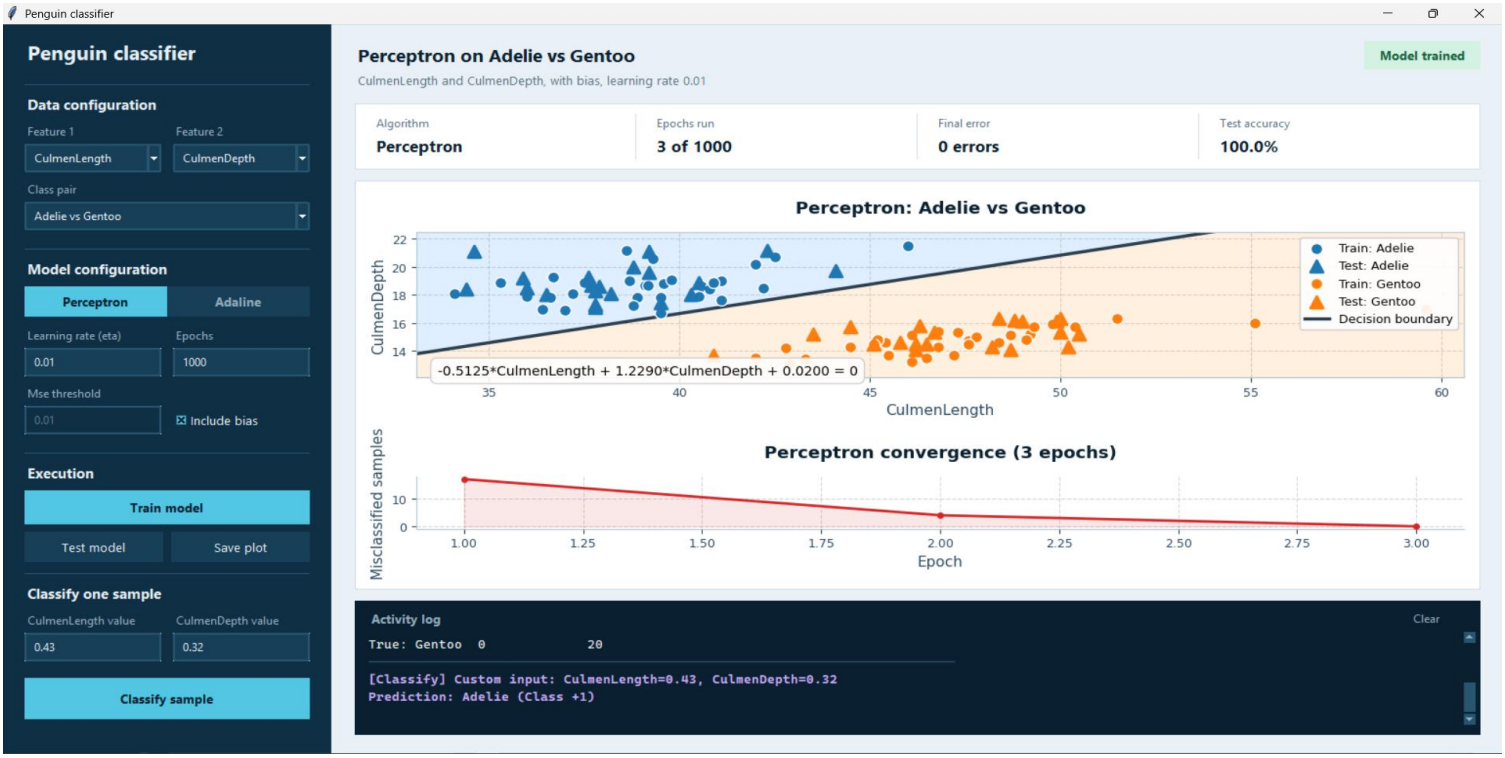
Test Case	Classes	Features Used	Data Quality	Perceptron Accuracy	Adaline Accuracy
Test 1	Adelie vs Chinstrap	CulmenDepth & BodyMass	**Bad Overlap**	**50.0%**	**60.0%**
Test 2	Adelie vs Chinstrap	CulmenDepth & FlipperLength	**Mediocre Overlap**	**52.5%**	**80.0%**
Test 3	Adelie vs Gentoo	CulmenLength & CulmenDepth	**Good Separable**	**100.0%**	**100.0%**
Test 4	Adelie vs Chinstrap	CulmenLength & CulmenDepth	**Good Separable**	**100.0%**	**100.0%**
Test 5	Chinstrap vs Gentoo	FlipperLength & BodyMass	**Good Separable**	**60.0%**	**100.0%**

Final Conclusion:

Based on the experimental results, the features that achieved the highest accuracy are **CulmenLength** and **CulmenDepth**, as they provided a perfectly linearly separable space (yielding 100% accuracy for both Perceptron and Adaline algorithms)

Our Gui

- ### Perceptron Testing



- ### Adaline Testing

